

Math 526, S06
Quiz 3

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Find a 3×3 matrix $\mathbf{A}$ so that, for any $3 \times n$ matrix $\mathbf{B}$ with rows $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$, (i) the first row of $\mathbf{AB}$ is $\mathbf{b}_1 + \mathbf{b}_2 - \mathbf{b}_3$, (ii) the second row of $\mathbf{AB}$ is $-2\mathbf{b}_2 + \mathbf{b}_3$, (iii) the third row of $\mathbf{AB}$ is $-4\mathbf{b}_1 + \mathbf{b}_2 + 3\mathbf{b}_3$.

Solution:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -2 & 1 \\ -4 & 1 & 3 \end{bmatrix}$$

Problem 2. (14 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization of $\mathbf{A}$.

Solution:

$$\begin{array}{ccc} \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} & \begin{bmatrix} 0 & 1 & -1 \\ 1 & -1 & 1 \\ -2 & 4 & -2 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} & \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ -2 & 4 & -2 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 0 & & \\ -2 & & \end{bmatrix} & \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 0 & & \\ -2 & 2 & \end{bmatrix} & \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ -2 & 2 & 1 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

b) Using the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization from (a) to solve $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Solution:

First step: Solve $\mathbf{Ly} = \mathbf{Pb}$. It is easy to find

$$\mathbf{Pb} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$$

So we solve

$$\begin{bmatrix} 1 & & \\ & 1 & \\ -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$$

We get

$$y_1 = 2, \quad y_2 = -3, \quad y_3 = 14.$$

Second step: Solve $\mathbf{Ux} = \mathbf{y}$.

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 14 \end{bmatrix}$$

We have

$$x_3 = 7, \quad x_2 = 4, \quad x_1 = -1.$$

So we have

$$\mathbf{x} = \begin{bmatrix} -1 \\ 4 \\ 7 \end{bmatrix}.$$